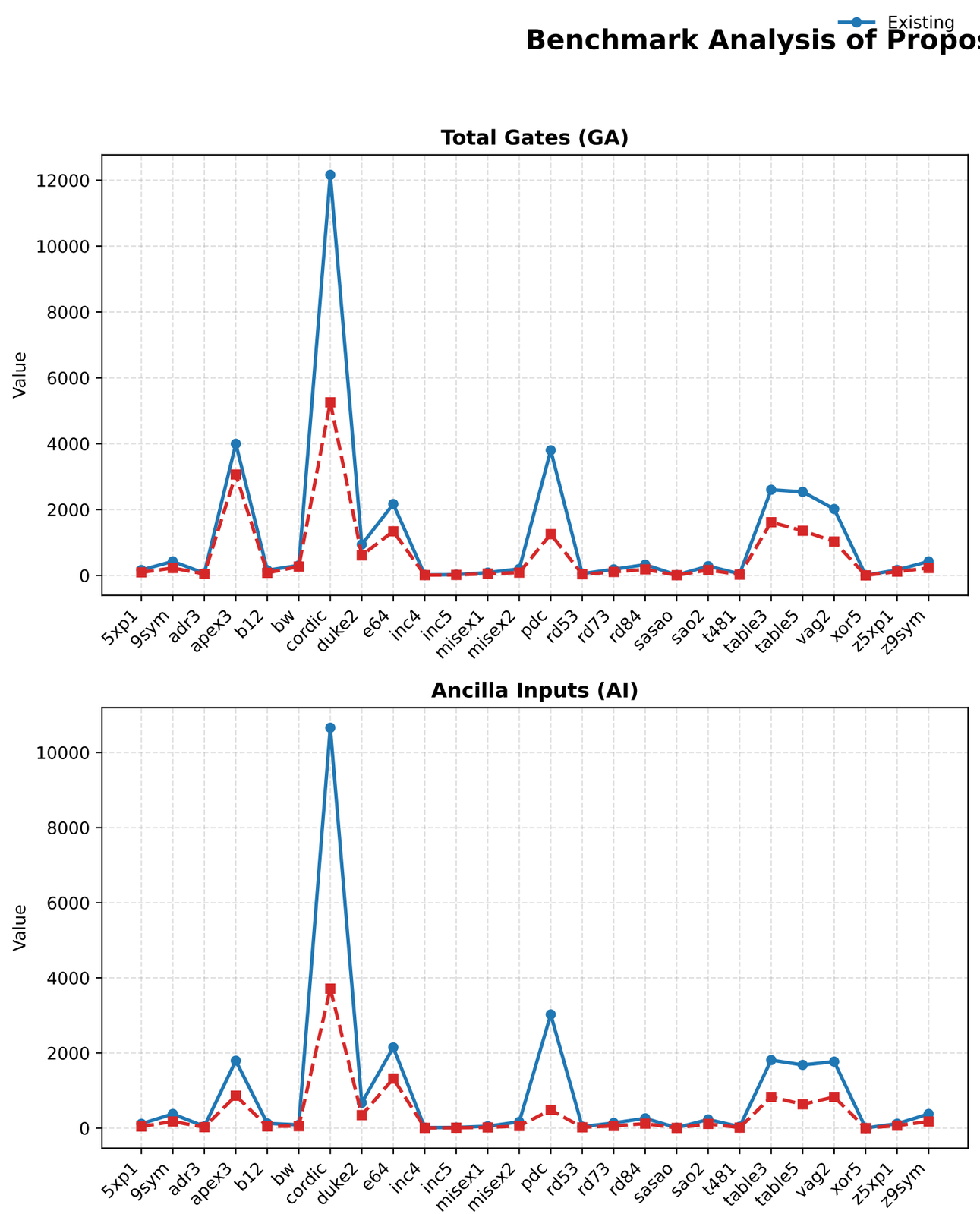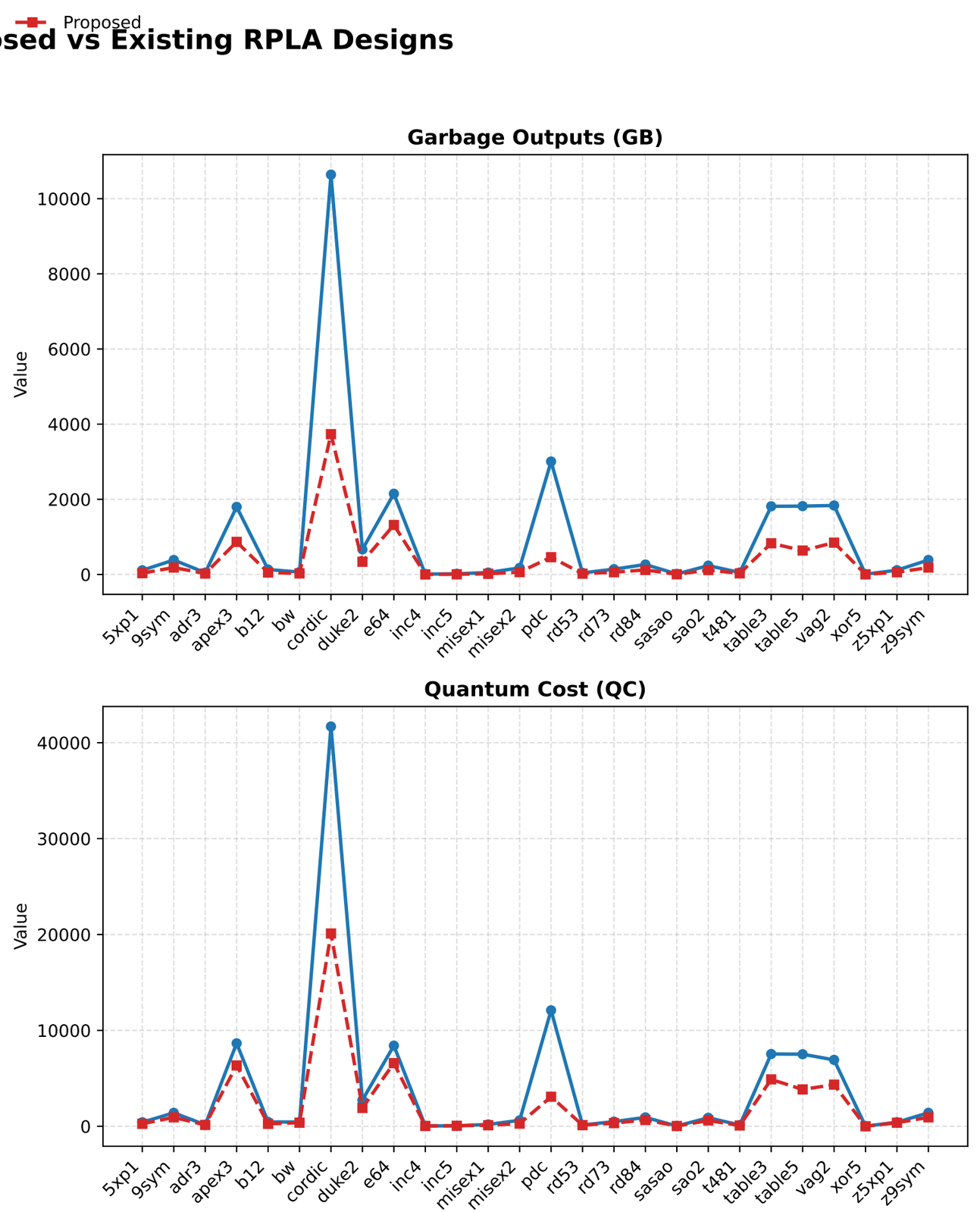Benchmark Analysis of Proposed vs Existing RPLA Designs